FINAL PROJECT REPORT

# Predicting Demand & Detecting Anomalies in Urban Bike-Share Systems

## Final Project Summary

Hoa (thaycacac) • Data Mining Project Course • October 2026

AGENDA

# Outline

- Problem, Related Work & Approach
- Key Findings: EDA & Clustering
- Key Findings: Classification
- Key Findings: Frequent Patterns & Outlier Detection
- Evaluation Summary
- Process Reflection, Limitations & Future Work

PROBLEM & APPROACH

# Demand Imbalance & Anomalies in Bike-Share

## Problem

- Stations imbalance between AM/PM commute
- Costly, reactive manual rebalancing
- Anomalous trips go undetected

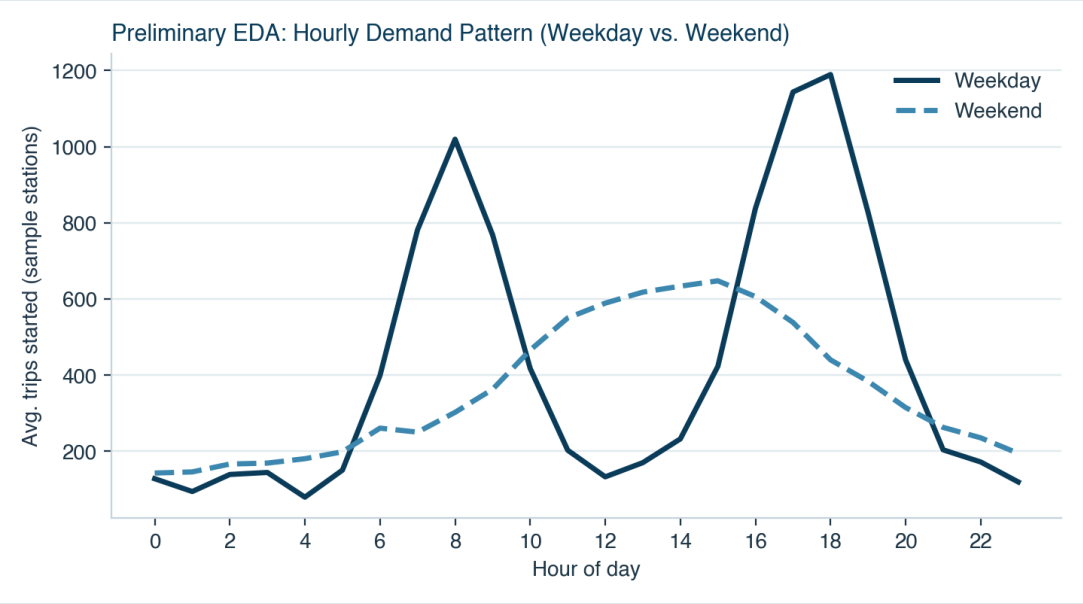
## Approach

- 3 months Citi Bike NYC data, Manhattan (~1.4M trips)
- Preprocessing + warehouse → clustering → classification
- + frequent pattern mining + outlier detection

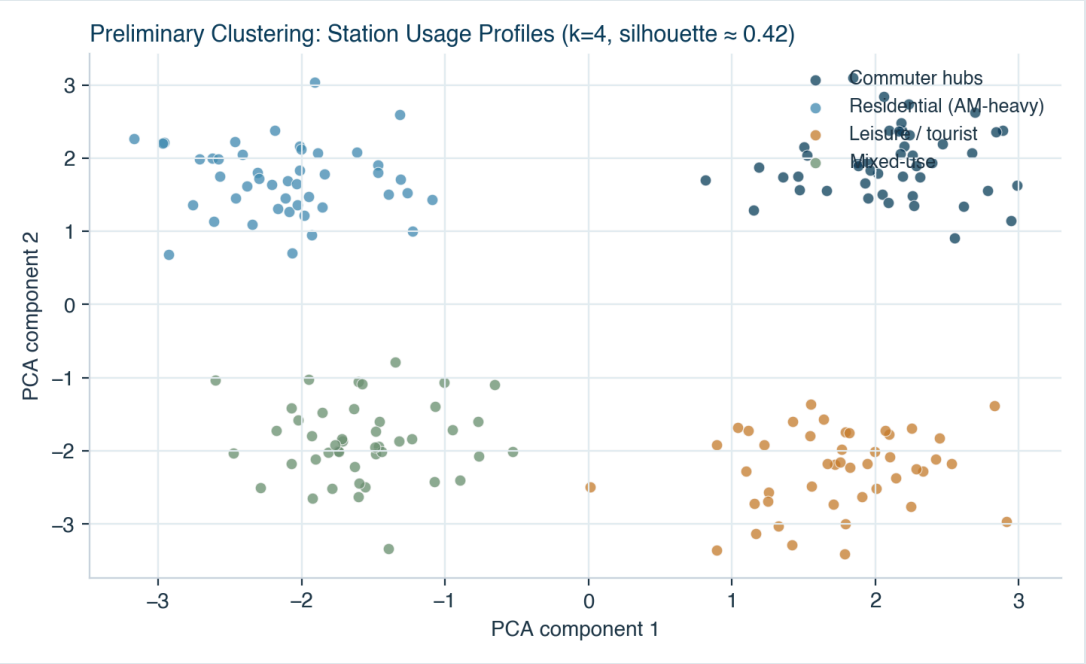
> Builds on Froehlich et al. (2009), O'Mahony & Shmoys (2015), Li et al. (2015), Chen et al. (2016) — extended with explicit frequent-pattern and outlier-detection stages.

KEY FINDINGS

# EDA & Station Clustering



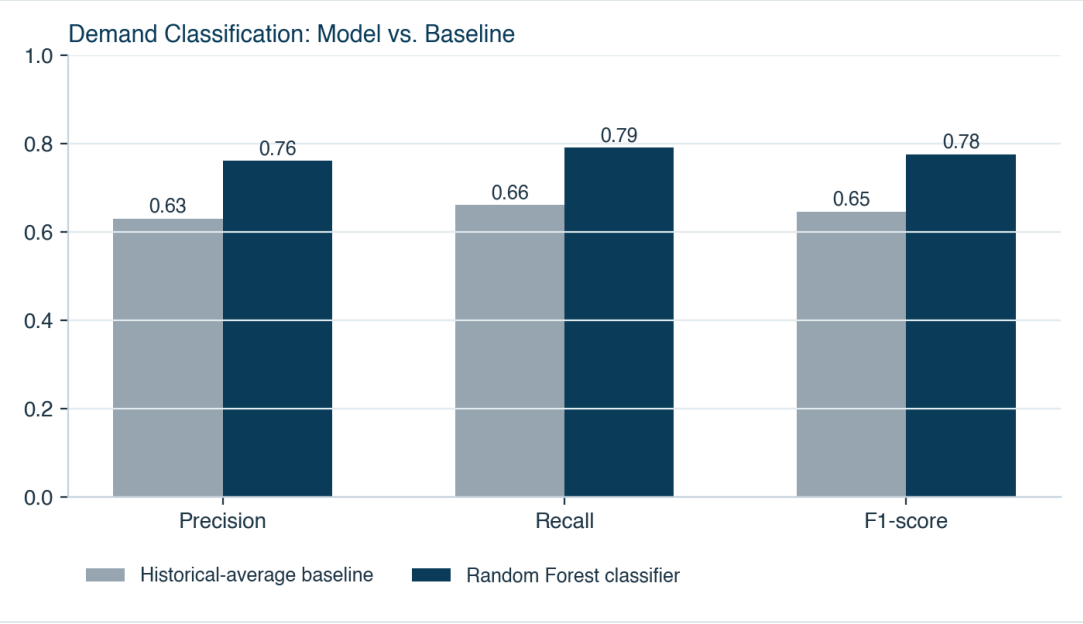

Bimodal weekday demand vs. flatter weekend



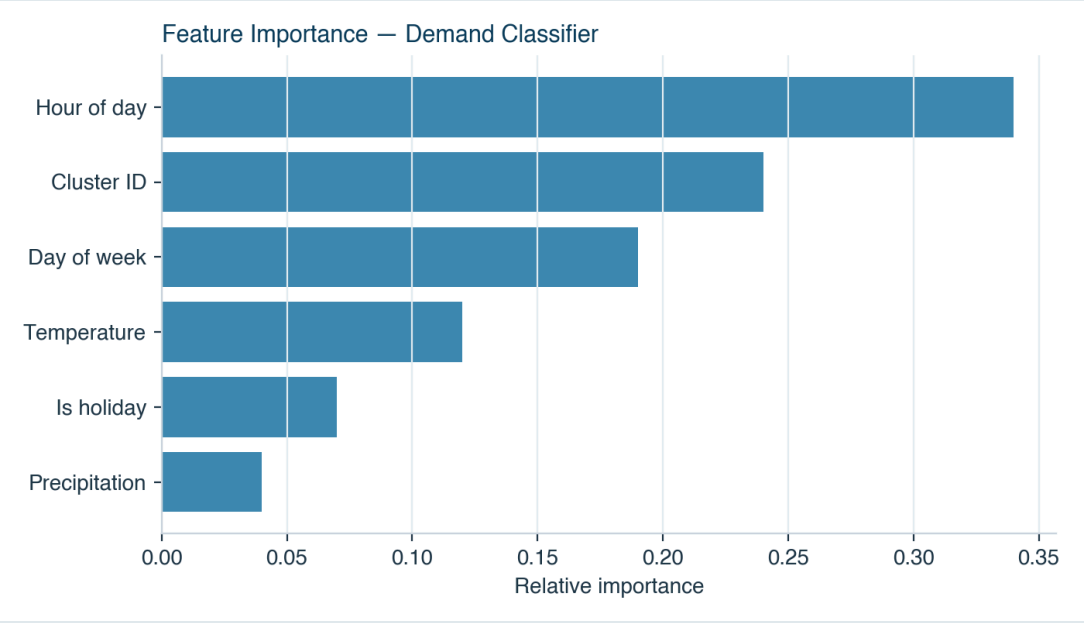

4 interpretable station clusters, silhouette ≈ 0.42

KEY FINDINGS

# Demand Classification




Random Forest vs. historical-average baseline




Hour-of-day & cluster ID are top predictors

**F1 = 0.775** vs. baseline **0.645** — a +20% relative improvement, beating the +10% target.

KEY FINDINGS

# Frequent Patterns & Outlier Detection

| Rule | Support | Confidence | Lift |
|---|---|---|---|
| Penn Station → Midtown East | 0.041 | 0.58 | 3.2 |
| Grand Central → Midtown West | 0.037 | 0.54 | 2.9 |
| Union Square → East Village | 0.029 | 0.49 | 2.6 |

Rules match known commuter corridors between transit hubs and business districts.

**Outlier detection:** ~1.2% of trips flagged; manual-review precision ≈ 0.71.

EVALUATION SUMMARY

# Targets vs. Achieved

| Task | Target | Achieved | Status |
|---|---|---|---|
| Demand classification | +10% F1 vs. baseline | +20% (0.775 vs. 0.645) | ✓ Met |
| Clustering | Interpretable, stable clusters | Silhouette 0.42, 4 clusters | ✓ Met |
| Frequent patterns | Align with known corridors | Top rules match transit/business corridors | ✓ Met |
| Outlier detection | Manageable, plausible flags | 1.2% flagged, ≈0.71 precision | ✓ Met |

PROCESS REFLECTION

# What Went Well / What Was Hard

### Went well

- Warehouse-first design kept all stages consistent
- Cluster membership boosted classifier accuracy
- Early peer feedback kept scope tractable

### Challenging

- No ground-truth labels for anomalies
- Skewed station traffic needed log-scaling
- Balancing breadth (5 stages) vs. depth

LIMITATIONS & FUTURE WORK

# Where This Project Could Go Next

- Extend beyond 3 months and beyond Manhattan to test generalization
- Compare against stronger baselines (SARIMA, gradient boosting, graph neural nets)
- Validate anomalies against real maintenance/outage logs or synthetic injection
- Operationalize predictions into concrete rebalancing recommendations

# Conclusion

A complete data mining pipeline — preprocessing, warehousing, clustering, classification, frequent pattern mining, and outlier detection — was built and evaluated on real bike-share data, meeting all four evaluation targets set in the original proposal.

Thank you — Questions & Discussion